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Real-Time Optimized Acid Stimulation of Oil Producer and Water Injector Wells Using Distributed Temperature Sensing (DTS), A Case Study Review

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Abstract

Recent acid stimulation practices involved the utilization of distributed temperature sensing (DTS) for optimizing the acid volumes and placement. This paper will provide an overview of two case study wells where DTS technology was deployed in acid stimulation jobs. The study aims to showcase the result of introducing DTS to acid stimulation in terms of treatment placement success rate, efficiency and cost. The process starts with the DTS generating an injection temperature variation profile enabling identification of thief zones including highly porous or fractured zones, and skin-damaged zones which typically require more acid treatment volumes. This in turn allows for accurate real-time acid placement resulting in an optimized treatment and maximizing the target zone productivity.

To stimulate or isolate specific interval zones, DTS technology can serve as an alternative to production logging tool (PLT) offering an innovative and efficient solution enhancing stimulation efficiency and enabling tailored acid stimulation using fit for purpose acid blend, placement and dosages. Two successful trials were conducted with an operator in the Middle East, one for an oil producer and the other for a water injector. The oil producer well stimulation using live coil DTS resulted in a 3-fold increase in production post-stimulation and a 1.5-fold increase in the maximum wellhead flowing pressure which meant a more sustainable production. The water injector well had a 2-fold increase in the maximum injection rate post-stimulation with a 68% wellhead injection pressure reduction which meant a more sustainable injection. These successful trials confirmed the effectiveness of live coil DTS technology in optimizing the stimulation treatment in both production and injection wells. Moreover, live coil DTS technology contributed to a more homogenous production/injection profile across the intervals of interest. By generating a temperature snapshot of the wellbore using a fiber optic installed inside the coiled tubing (CT) string, the DTS significantly reduced operational cost and helped maximize the productivity potential from the stimulation treatment. Additionally, DTS technology enabled real-time interpretation while CT is at the target zones which facilitated immediate acid placement optimization.

This innovative and pioneering technology minimized the time required for well evaluation at offshore sites by approximately 90%. This paper will provide a case study review for two high-profile stimulation jobs in which both live coil and DTS technologies were used together to maximize the acid treatment placement accuracy and treatment efficiency and success. The addition of DTS also allowed for a remarkable enhancement of the overall stimulation treatment by allowing real time interpretation and decision making.

Introduction

The evolution of data acquisition technologies in non-instrumented wells has significantly transformed the way stimulation treatments and production are diagnosed and optimized in real time. In this context, distributed fiber-optic sensing services—specifically distributed temperature sensing (DTS) and distributed acoustic sensing (DAS) conveyed through coiled tubing (CT), have emerged as a highly effective and versatile alternative to conventional well logging. This technology enables continuous measurement of thermal and acoustic parameters across the entire wellbore, providing real-time insight during planned interventions.

The Real-Time Fiber Optic Coiled Tubing (RTFOCT) system integrates a 4-mm outer diameter fiber-optic cable and data housed within a CT String (as shown in [Figure 1](#), equipped with 6 optic fibers: fiber-optic cable includes a single-mode fiber for DAS, five multi-mode fibers for DTS and telemetry, enabling real-time transmission from downhole sensors and cameras - if equipped- in addition to three back up fibers). Surface acquisition equipment is used for real-time data processing and visualization and may be equipped with an optional modular bottomhole assembly (sensing BHA can be configured to measure internal and external pressure, temperature, depth using a casing collar locator, gamma ray, tension/compression and torque). This setup allows for the evaluation of production or injection profiles, leak detection, cluster efficiency analysis, monitoring of chemical and acid treatments and well communication within a single CT run. ([Al Matar et al. 2020](#))



Figure 1—CT-conveyed fiber optic cable deployed inside a CT string ([Ellmauthaler et al. 2019](#))

Surface systems receive and process this data through a slip ring assembly mounted on the CT reel, ensuring uninterrupted power supply and continuous data flow. A data acquisition system, DTS/DAS junction boxes, and interrogator units enable remote visualization and global transmission via satellite, allowing subject matter experts and decision-makers to respond immediately to downhole events. The reel termination assembly (RTA) anchors the fiber cable while enabling pumping continuity. At the surface, DTS boxes and acquisition units are pre-terminated and configured to facilitate fast deployment and immediate data streaming.

In stimulation operations, RTFOCT technology proves particularly valuable. It enables engineers to assess fluid distribution across the perforated interval using DTS, determine high and low intake zones. The effectiveness of acid and chemical treatments can be qualitatively analyzed by interpreting DTS thermal profiles, helping operators fine-tune stimulation designs on the fly. Additionally, fracture gradient and bottomhole breakdown pressures can be estimated, providing critical insights for future intervention planning.

Recent developments in RTFOCT include the integration of thermo-hydraulic-mechanical modeling to provide quantitative assessments. These models combine mass balance (continuity), momentum (equation of motion) and energy equations (based on temperature) for the wellbore fluid and similar balances for the reservoir domain. A unique aspect of the model is that it accounts for fluid losses into the formation, which influence the wellbore pressure profile beyond hydrostatic and frictional considerations. This allows for improved prediction of inflow performance and injection efficiency.

In real-world applications, RTFOCT has demonstrated its ability to replace multiple conventional intervention tools—such as wireline and electric CT—by offering integrated diagnostic and treatment evaluation in one single trip. For instance, bullheading treatments in highly heterogeneous formations often result in uneven fluid placement. By contrast, RTFOCT enables selective placement using real-time diagnostics to prevent overtreatment of highly permeable zones. Moreover, the use of fiber-optic CT services contributes to reducing non-productive time and operational costs. Real-time monitoring enables smarter decision-making, leading to better reservoir management and enhanced treatment outcomes. Studies have shown that the added value of RTFOCT in stimulation campaigns justifies its implementation in both new and mature wells. In summary, fiber-optic distributed sensing conveyed by coiled tubing is reshaping intervention workflows by enabling continuous, real-time diagnostics along the entire wellbore. The RTFOCT system offers a single-platform solution for data acquisition, treatment execution and performance evaluation, making it an essential tool for modern well intervention strategies. (Butz et al. 2020)

DTS Workflow

In the context of stimulation diagnostic operations involving Distributed Temperature Sensing (DTS) understanding the temporal evolution of temperature profiles is critical. Among the most fundamental and commonly used phases during such operations are the baseline, injection, and warmback periods. Each of these stages plays a unique role in data interpretation and reservoir characterization. It is a common practice to execute the three stages prior and after the stimulation treatment is pumped in the well in order to evaluate the changes in profile and subsequently the effectiveness of the stimulation treatment. (Santin et al. 2018)

Baseline

The baseline period represents the initial phase of the operation, during which the well remains in a static or undisturbed condition prior to any fluid injection. During this time, the DTS system records the native geothermal gradient and establishes a reference thermal profile along the entire wellbore. This reference is essential for comparison with subsequent thermal responses induced by fluid movement during injection and post-injection (warmback) phases. Baseline measurements allow engineers to detect subtle anomalies in the temperature distribution that may indicate pre-existing wellbore issues such as leaks, crossflows or fluid communication between zones. Furthermore, establishing an accurate baseline is crucial for isolating temperature changes solely attributable to the injection process, thereby improving the reliability of subsequent analysis. (McCarthy et al. 2020)

Injection

The injection phase begins once treatment fluids are pumped into the wellbore. During this stage, the DTS system captures thermal responses associated with the fluid entering the formation. The thermal footprint of the injection is typically characterized by a localized cooling effect due to the lower temperature of the injected fluid compared to the reservoir. By analyzing the temperature drop over time and depth, it is possible to identify high-intake zones, flow distribution across perforations and potential thief zones. Real-time monitoring during this phase enables optimization of the injection schedule and pump rate, ensuring effective placement of stimulation or chemical treatments.

Warmback

Following the cessation of injection, the system enters the warmback period. This stage involves the gradual return of the wellbore temperature toward the baseline geothermal profile as the formation and wellbore exchange thermal energy. The rate and pattern of this warmback response carry significant diagnostic value. Zones with higher permeability or greater fluid intake typically exhibit slower warmback recovery due to continued fluid invasion or the presence of residual cool fluids. In contrast, low-permeability zones or poorly stimulated intervals may exhibit a faster temperature recovery. The interpretation of the warmback phase allows for the evaluation of the effectiveness of the injection, identification of heterogeneities in reservoir response and the inference of the near-wellbore properties such as fracture conductivity, flow capacity and thermal diffusivity. When modeled quantitatively using thermo-hydraulic simulations, warmback profiles can provide estimates of fracture geometry and reservoir contact area.

Methodology & Results

Two candidates (an oil producer and a water injector) were stimulated using DTS technology to optimize the acid placement and volumes. The execution of each candidate along with the treatment optimization is summarized below.

Candidate A

Candidate A is an oil producer that was drilled and completed in Q4 of 2023. The upper completion is equipped with an electric submersible pump (ESP), and the lower completion is equipped with 8 swellable packers and 11 inflow control devices (ICD) composing 7 layers as per [Table 1](#) below.

Table 1—ICD sizes and distribution

Layer	Top (ft)	Bottom (ft)	ICD	Depth (ft)	ICD	Depth (ft)
1	12,264	12,462	1: 4 mm x 4	12,335	2: 4 mm x 4	12,403
2	12,474	12,618	3: 2.5 mm x 4	12,525	4: 2.5 mm x 4	12,591
3	12,630	12,720	5: 4 mm x 4	12,661		
4	12,733	12,931	6: 2.5 mm x 4 & 1.6-mm x 4	12,806	7: 2.5 mm x 4 & 1.6-mm x 4	12,872
5	12,944	13,082	8: 2.5 mm x 4	12,999	9: 2.5 mm x 4	13,064
6	13,094	13,338	10: 4 mm x 4	13,196		
7	13,350	13,445	11: 1.6 mm x 8	13,380		

Temperature data was collected during the pre-stimulation injection period, to identify the zones of high and low permeability. During the pre-stimulation injection, it was observed that the deepest layer (layer 7) appeared to take the least amount of fluid compared to the rest of the layers. In addition, minimal cooling was observed in layer 7 during the injection period. Cooling was observed in layers 1-6 during the cooldown period. The temperature data was then collected during the warm back period for all 7 layers. The analysis of the warm back period temperature profile showed that layer 7's temperature profile did not change. This indicated that layer 7 was not taking any fluid. The rest of the layers had a visible increase in the temperature from the temperatures recorded during cooldown. This meant that all the layers were taking fluid except for

layer 7. As a result, the stimulation treatment pumping schedule was optimized to inject the highest dosage of gelled acid in layer 7 and a smaller dosage in the rest of the layers. The gelled self-diverting acid was used due to the heterogeneous permeability profile, since it will block the high permeability zones and divert the acid to penetrate the lower permeability areas, thus ensuring a homogenous profile. The temperatures observed during the post-stimulation injection and temperature rebound test showed that cooling was observed in layers 6 and 7. Post stimulation confirmed the effectiveness of the stimulation treatment design showing a more uniform warm back throughout layer 1 to 6 and therefore an even connection between the reservoir and the completion was achieved. Layer 7 due to, potentially, its own petrophysics properties as well as limited entry through the ICD (smallest openings) remains mostly unconnected, however layer 6 is now fully opened to production from heel to toe. This can also be observed in [Figure 2](#) where the % of fluid leaving the wellbore in layer 6 increased from 23% to 51%. This is significant considering the depth of layer 6 and that a large portion of it was receiving a minimal amount of injected fluid during the cooldown phase.

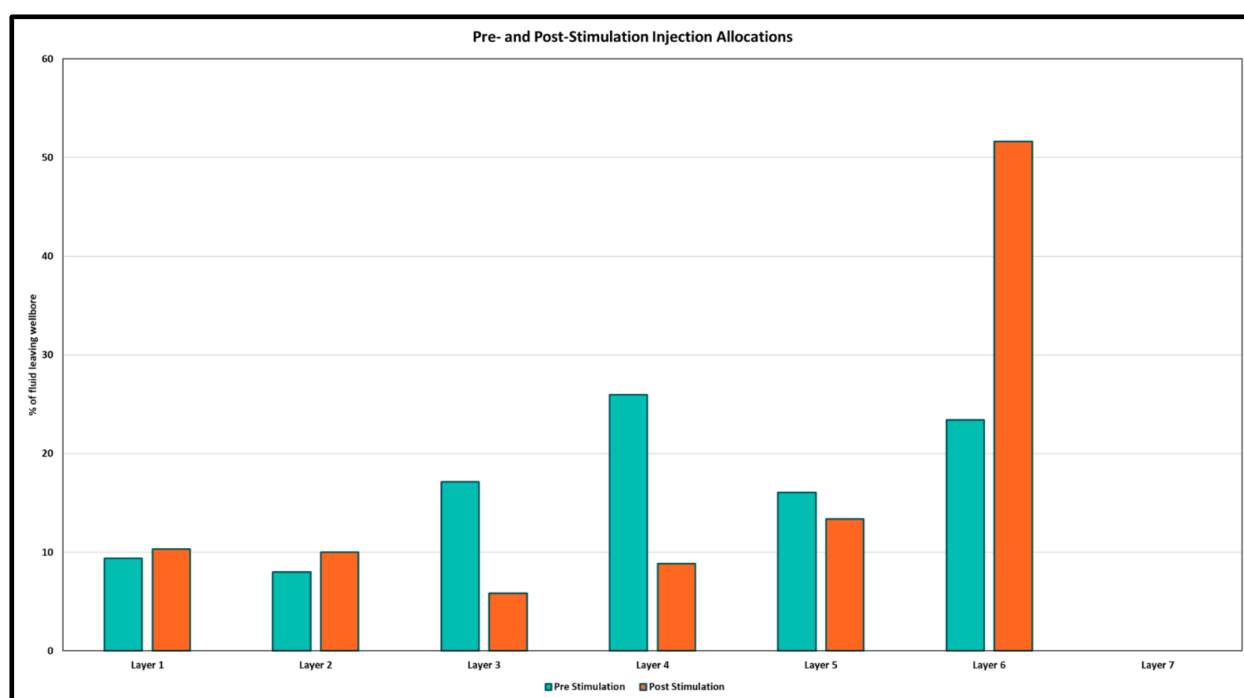


Figure 2—Pre and post stimulation injection allocations from temperature build up data

Stimulation of candidate A resulted in a 3-fold increase in the maximum flow rate per day with a 1.5-fold increase in maximum wellhead flowing pressure. Additionally, using the live coil DTS technology process led to a more homogenous production profile as observed in [Figure 3](#).

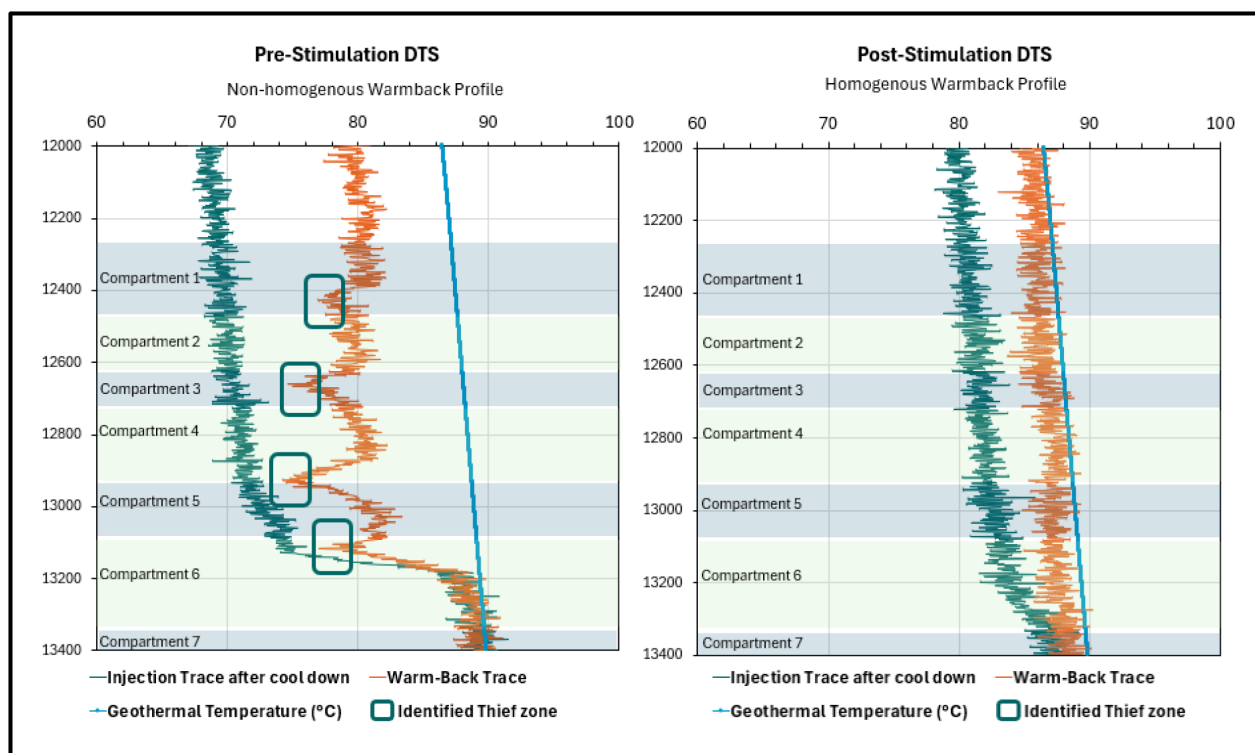


Figure 3—Pre vs post stimulation DTS profiles for Candidate A

Candidate B

Candidate B is a water injector that was drilled with a lower completion equipped with 6 swellable packers and 7 pre-perforated liners (PPL) consisting of the 6 layers in Table 2 below.

Table 2—Zone and layer identifications

Layer	Top	Bottom	Zone
1	14,965 ft	15,208 ft	C1
2	15,208 ft	15,537 ft	C2
3	15,552 ft	15,777 ft	C3
4	15,792 ft	17,550 ft	C5
5	17,566 ft	18,400 ft	C7
6	18,415 ft	18,641 ft	C9

During the pre-stimulation injection, it was observed that the injected fluid did not travel past 17,430 ft and that minimal cooling was observed in the well below this depth (as observed in Figure 4). This meant that the injected fluid did not reach the C7 and C9 layers. It is possible that the deepest two intervals were taking fluid but that the injected fluid front had not reached those zones by the end of the injection period. Since this is a horizontal well, cooling in the toe section of the well will not occur until all the original

wellbore fluid in the horizontal section of the well has been displaced into the formation and injected fluid or fluid from the overburden section has reached the area of interest.

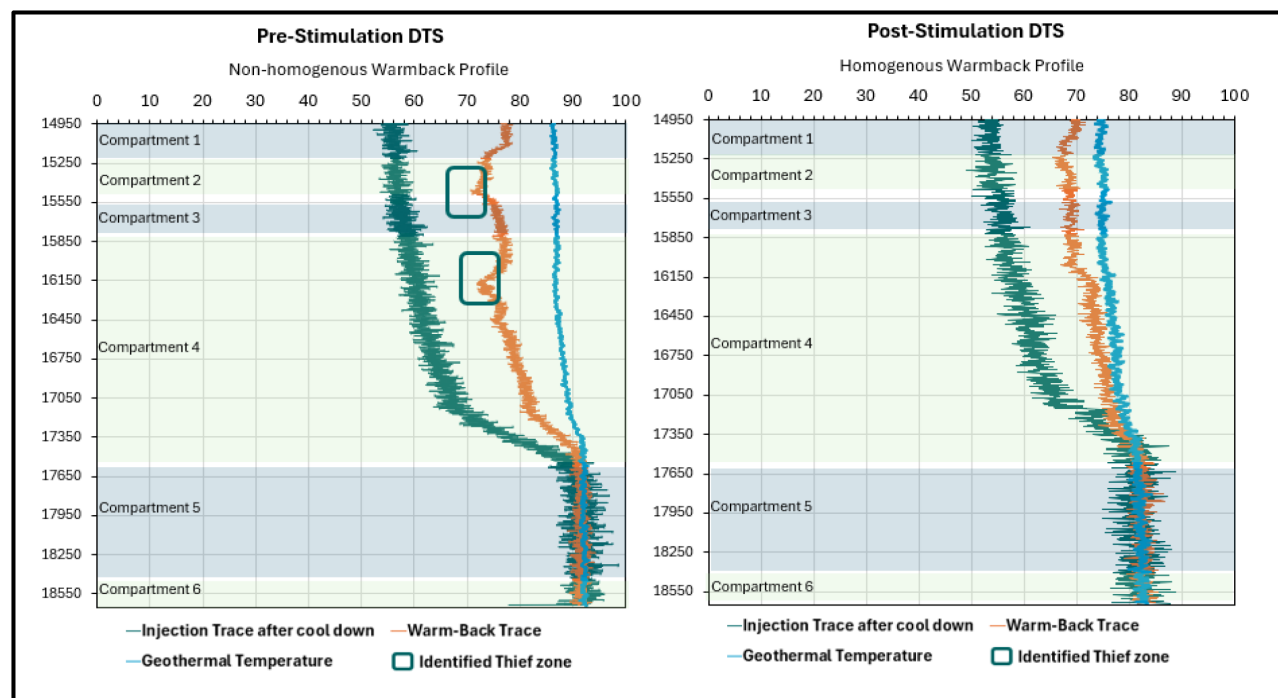


Figure 4—Pre vs post stimulation DTS profiles for Candidate B

It is likely that the injected volume and rate was insufficient for injected fluid to reach swell packer 5. Looking at the temperature profile it appears that, when injection was shut down, temperatures were still declining at 17,430 ft suggesting that injected fluid continued to be displaced downhole. But since no temperature changes were observed below 17,430 ft, the injection was limited to the upper 4 injection intervals. The warm back temperatures during the pre-stimulation injection also confirm that minimal injected fluid went deeper than layer 5.

The optimized stimulation pumping schedule can be seen in Table 3. The well was stimulated by pumping diesel + 10% mutual solvent (M.S) for pre-flush followed by retarded acid which allows for a deep acid penetration into the low permeability zones. The retarded acid (RET) was coupled with a viscoelastic acid (VES) for diversion from the high permeability zones to the lower permeability zones. A larger retarded acid volume and higher dosage was selected to cover layers 5 and 6 in order to ensure an improvement in well performance and production. The remaining retarded acid volume was distributed across the remaining layers based on the permeability and behavior observed during the cooldown and warmback. The layers that cooled down the most and warmed back the least were considered to be highly permeable. These layers were given a smaller dosage of retarded acid and a higher dosage of diverting viscoelastic acid. In a similar manner, the layers that were the least permeable and warmed back quickly, had a higher dosage of retarded acid. During the pre-stimulation injection, the maximum rate achieved was 6 bbl/min at a wellhead pressure (WHP) of 2,000 psi. The post-stimulation injection results showed that there was a 2-fold increase in maximum injection rate per day with a ~68% wellhead injection pressure reduction. In addition to this enhancement, there was a more uniform warm back observed indicating a more homogenous injection post-stimulation.

Table 3—Optimized stimulation pumping schedule for candidate B

Step	Step Type	Step Time hh:mm:ss	Elapsed Time hh:mm:ss	Fluid Type	Fluid Vol (bbl)	Flow Rate (bpm)	CT From (ft)	CT To (ft)	# passes	CT Speed (ft/min)
1	Pre-Flush	0:50:00	0:50:00	Diesel + 10% M.S	100	1	14,929	18,550	1	37
2	Main acid	7:35:30	8:25:30	20% RET	911	2	18,550	17,543	3	7
3	Main acid	1:42:05	10:07:35	20% RET	204	2	17,543	17,200	3	10
* Flushing stage: POOH above WLEG while pumping 300 bbl of FW- First 75 through CT only then open CT/Tbg annulus and continue flushing stage - then RIH and continue pumping acid stage										
4	Diversion	1:50:23	11:57:58	20% VES	221	2	17,200	16,041	3	32
5	Main acid	3:38:00	15:35:58	20% RET	436	2			3	16
6	Main acid	1:35:00	17:10:58	20% RET	190	2	16,041	15,775	3	8
7	Main acid	1:28:56	18:39:54	20% RET	178	2	15,775	15,526	3	8
8	Diversion	1:46:36	20:26:30	20% VES	213	2	15,526	14,929	3	17
9	Post-Flush	0:50:00	21:16:30	Diesel + 10% M.S	100	2	14,929	14,929	N/A	N/A
10	Displacement	3:45:00	25:01:30	Diesel	450	2	14,929	14,929	N/A	N/A
11	Displacement	0:50:00	25:51:30	Nitrified Diesel	100	2	14,929	0	N/A	N/A

Conclusions

The traditional approach for evaluating the reservoir injection/production contribution is time consuming and inefficient. In addition, the additional human effort and operational steps introduce potential safety hazards and risks. The use of live coil DTS technology enabled the evaluation of the injection/production contribution while performing the acid stimulation with the same coiled tubing reel. Live coil DTS technology was deployed to successfully, efficiently and safely stimulate two wells, an oil producer and a water injector. The oil producer stimulation resulted in a 3-fold increase in the maximum flow rate per day with a 1.5-fold increase in maximum wellhead flowing pressure. In addition, the water injector stimulation achieved a 2-fold increase in maximum injection rate per day with a 68% reduction in wellhead injection pressure. The outcome resulted from effective acid treatment placement and real-time interpretation and optimization facilitated by DTS technology. DTS technology can also be used for different applications such as ICD shifting, which can result in an improvement in oil/gas production or mitigation of unwanted fluid production.

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Nomenclature

- BHA - Bottomhole Assembly
- CT - Coiled Tubing
- DAS - Distributed Acoustic Sensing
- DTS - Distributed Temperature Sensing

ESP - Electric Submersible Pump
ICD - Inflow Control Devices
MS - Mutual Solvent
PLT - Production Logging Tool
PPL - Pre-Perforated Liner
RET - Retarded Acid
RTA - Real Termination Assembly
RTFOCT - Realtime Fiber-Optic Coiled Tubing
VES - Viscoelastic Acid
WHP - Wellhead Pressure

References

- Al Matar M., Mohapatra S., Al-Ateeqi H. et al 2020. *Coiled Tubing-Assisted Real-Time Temperature Mapping for Discerning Completion Integrity Followed by Optimized Placement and Evaluation of a Porosity Fill Treatment for Water Shutoff in Dual-String Wells: Customized Solution*. Presented at SPE/ICoTA Well Intervention Conference and Exhibition, Houston, Texas, USA 24-25 March. <https://doi.org/10.2118/199804-MS>
- Butz S, Williams T and McCarthy C. 2020. *Using Simultaneous Services to Provide Operational Efficiency Gains and Reduce Operational Costs for Coiled Tubing/ Fiber Diagnostic Services*, Presented at SPE/ICoTA Well Intervention Conference and Exhibition, Houston, Texas, USA 24-25 March. <https://doi.org/10.2118/199789-MS>
- Ellmauthaler A., Palacios W., LeBlanc M. et al 2019. *Vertical Seismic Profiling via Coiled Tubing-Conveyed Distributed Acoustic Sensing*. Presented at SPE/ICoTA Well Intervention Conference and Exhibition, Houston, Texas, USA 26-27 March. <https://doi.org/10.2118/194244-MS>
- McCarthy C, Williams T and Butz S. 2020. *High Accuracy Leak Detection Performed on Entire Live Wellbore in RealTime Using Fiber-Optic Diagnostic-Capable Coiled Tubing*. Presented at: International Petroleum Technology Conference held in Dhahran, Saudi Arabia, 13 – 15 January. <https://doi.org/10.2523/IPTC-19576-MS>
- Santin Y., Matar K., Montes E. et al, 2018. *Reliable Carbonate Stimulation Using Distributed Temperature Sensing Diagnostics and Real-Time Fiber-Optic Coiled Tubing Intervention in Kuwait*. Presented at: SPE Asia Pacific Oil & Gas Conference and Exhibition held in Brisbane, Australia, 23–25 October. <https://doi.org/10.2118/191914-MS>